

Axioms of Probability

1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$. This can be extended to $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for mutually exclusive events $A_1, A_2, \dots, A_n$.

Basic Properties of Probability:

$$\forall A \in S, P(A) < 1$$

$$P(A^c) = 1 - P(A)$$

$$P(\emptyset) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \subseteq B \implies P(A) \leq P(B)$$

Counting:

Permutation — An arrangement of objects in a specific order. $P(n, r) = \frac{n!}{(n-r)!}$

Combination — A selection of objects without regard to the order. $C(n, r) = \frac{n!}{r!(n-r)!}$

Binomial Coefficient — The number of ways to choose k successes in n independent Bernoulli trials. $\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$

Number of Subsets — The number of subsets of a set with n elements is 2^n .

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication Rule: — $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

$$P(\bigcap_{i=1}^n A_i) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|\bigcap_{i=1}^{n-1} A_i)$$

Law of Total Probability: — If $B_1, B_2, \dots, B_n$ are mutually exclusive and exhaustive events, then for any event A : $P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$

Bayes' Theorem: —

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

Independence:

Two events A and B are independent if and only if $P(A|B) = P(A)$ and $P(A \cap B) = P(A)P(B)$.

If A and B are independent, then B and A , A and B^c , A^c and B , A^c and B^c are also independent.

Random Variables An RV is a function that assigns a real number to each outcome in the sample space of a random experiment.

Discrete RV — A random variable that can take on a countable number of distinct values.

Continuous RV — A random variable that can take on any value within a given range or interval.

$X : S \rightarrow \mathbb{R}$ Where S is the event space and X is the random variable.

PMF — $f(x) = P(X = x)$ (discrete)

$$P(a \leq X \leq b) = \int_a^b f(x)dx \text{ (continuous)}$$

CDF — $F(x) = P(X \leq x) = \sum_{y \leq x} P(X = y)$ (discrete)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt \text{ (continuous)}$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

Bernoulli Random Variable — A discrete RV that takes the value 1 with probability p and the value 0 with probability $1 - p$.

Expected Value — $E(X) = \mu_X = \sum_x xP(X = x)$ or $= \int_{-\infty}^{\infty} xf(x)dx$

$$E(h(X)) = \sum_x h(x)P(X = x) \text{ or } E(h(X)) = \int_{-\infty}^{\infty} h(x)f(x)dx$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(X^2) = \sum_x x^2 p(x) \text{ or } = \int x^2 f(x)dx$$

Variance — $\sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - (E(X))^2$ or $\sigma_X^2 = \int (x - \mu_X)^2 f(x)dx$

Standard Deviation — $\sigma_X = \sqrt{\text{Var}(X)}$